



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS

Software Requirement Specification For
Support Ticket Management Module

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Abbreviations

Abbreviation	Full Form
SRS	Software Requirements Specification
CR	Class Representative
HOD	Head of Department
SLA	Service Level Agreement
RBAC	Role-Based Access Control
DFD	Data Flow Diagram
REST	Representational State Transfer
SMTP	Simple Mail Transfer Protocol
CI/CD	Continuous Integration / Continuous Deployment

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Support Ticket System to be built for Pulchowk Engineering Campus. The system will let students, class representatives (CRs), department heads (HODs), the Academic Section, the Financial Team, and the CIT team manage support requests through a simple ticketing workflow. This SRS covers both functional and non-functional requirements and is the primary reference for development, testing, and deployment.

1.2 Document Conventions

Convention	Meaning
TKT-NNN	Ticket/backlog item identifier
Must / Shall	Mandatory requirement
Should	Recommended requirement
May	Optional requirement
[FT-N]	Functional requirement number
[NFR-N]	Non-functional requirement number

1.3 Intended Readers and Stakeholders

This document is intended for the following readers and stakeholders:

Student — Submits tickets for personal academic and facility issues. Can view, reply to, and close their own tickets. Can reopen closed tickets within 30 days if the issue persists.

CR (Class Representative) — Acts on behalf of an entire class section. Can submit class-level tickets with auto-filled class/section. Inherits all Student capabilities.

Staff — Handles tickets assigned to their department based on category. Can add internal notes, change ticket status, reply in thread, and escalate to Dept Admin if needed. Receives notifications for assigned tickets.

Dept Admin (HOD — Head of Department) — Manages all department-level tickets. Can assign, reassign tickets to staff within the department. Receives escalated tickets from staff. Can escalate tickets to Campus Admin when the issue is beyond departmental authority. Views department dashboard and statistics.

Campus Admin (System Admin) — Has full system access across all departments. Manages routing rules, SLA deadlines, users, and roles. Handles tickets escalated from Dept Admins. Configures notification templates. Views all reports and dashboards.

1.4 Product Scope

The Support Ticket System is a web-based module that will be integrated into CIT's existing campus management system. It replaces the current informal practice where students email random staff members or visit offices in person for issues like internet problems, fee queries, grade disputes, lab equipment requests, and departmental complaints.

The system covers:

- Ticket submission and tracking (students and CRs)
- Automatic routing based on category (CIT, Academic, Finance, HOD, etc.)
- Internal notes for staff (hidden from students)
- Notifications via email and in-app
- Escalation when deadlines are missed
- Dashboards and reports for HODs and admins

The system serves approximately 4,000+ students across six departments: Civil Engineering, Electrical Engineering, Computer Engineering, Mechanical Engineering, Architecture, and Applied Sciences. CIT already maintains the network, the campus portal, and the student database — this module will hook into those existing systems.

2. Overall Description

2.1 Product Perspective

The Support Ticket System is not a standalone application. It is a module that lives inside CIT's existing campus management system. This means:

- Authentication comes from the existing campus login system (students already have accounts)
- User data (name, department, section, role) is pulled from the existing student/staff database
- The UI will be embedded as a new section within the existing campus portal
- No separate login page — users access it from their existing dashboard

The CIT team manages the parent system. Our team develops the ticket module on a separate branch. Once we complete a sprint's work, the CIT team reviews the code and merges it into the main system. This branch-and-review workflow is the agreed collaboration model.

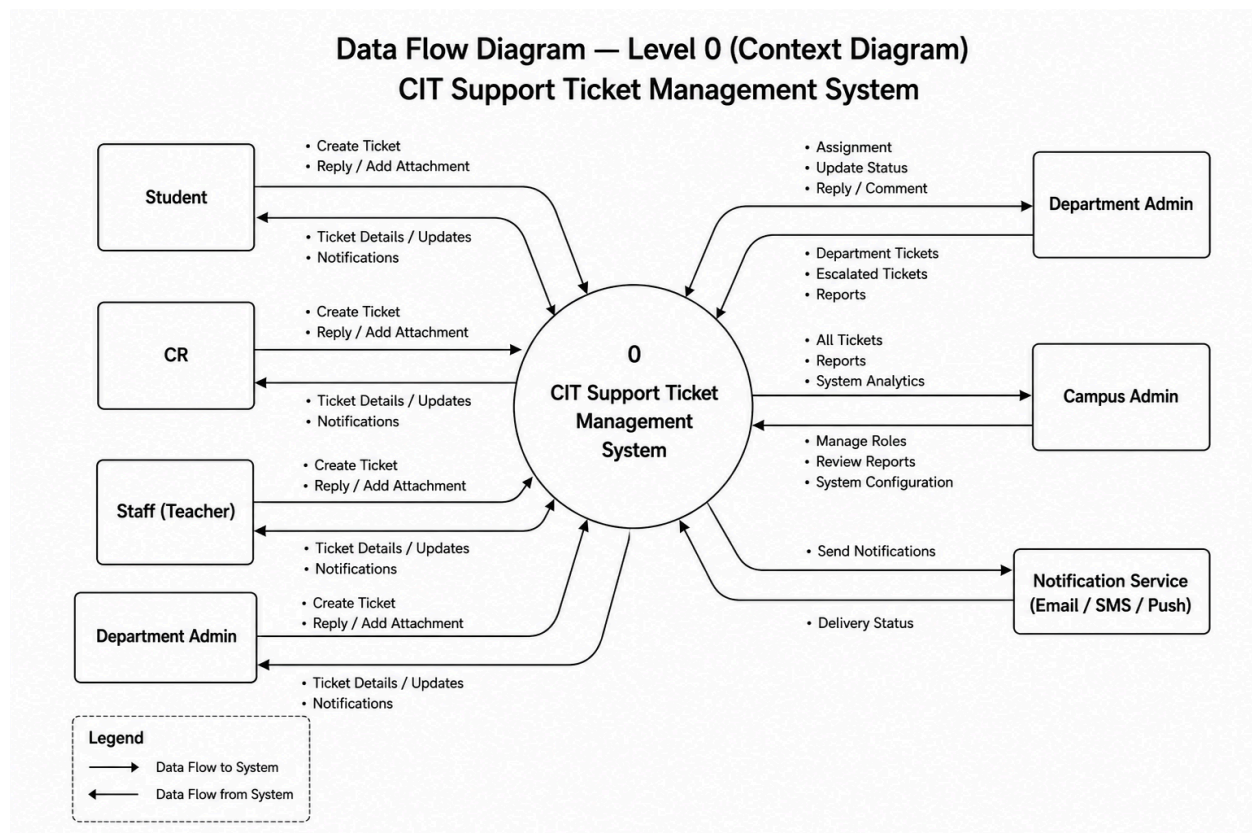


Figure 1: Data Flow Diagram — Level 0 (Context Diagram)

2.2 Product Functions (Summary)

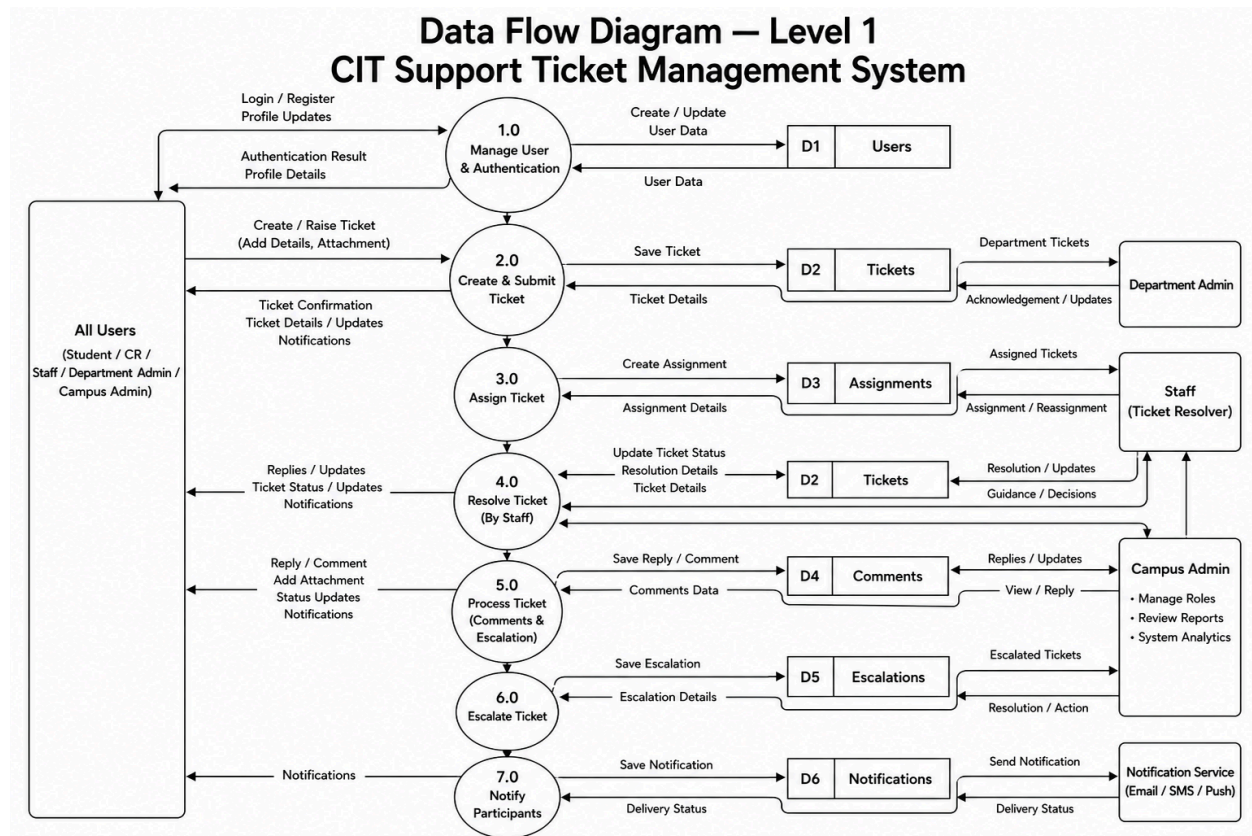


Figure 2: Data Flow Diagram — Level 1 (Process Decomposition)

Area	What the system does
Ticket Creation	Students and CRs submit tickets with title, description, category, attachments
Routing	Incoming tickets are automatically categorized and assigned to the appropriate department/staff based on routing rules
Thread View	Each ticket has a conversation thread with messages, files, and internal notes
Notifications	Email and in-app notifications for assignments, replies, and status changes
Escalation	Unattended or overdue tickets escalate: Staff → Dept Admin → Campus Admin
User Management	Admins can add/manage users and roles; bulk import from Excel
CR Features	CRs can submit tickets on behalf of their entire class section
HOD Dashboard	Dept Admins see department tickets, can reassign, escalate, view stats
Admin Panel	Routing rules configuration, SLA deadlines, user/role management, notification templates
Search	Full-text search across ticket title, ID, description, and messages
Reports	Dashboards with charts, team performance metrics, Excel export

2.3 User Characteristics and Roles

Student

- Submits tickets for personal issues (internet, fees, grades, etc.)
- Can view and reply to their own tickets
- Cannot see internal notes added by staff
- Cannot see other students' tickets

- Can reopen a closed ticket within 30 days of closing if the issue is not resolved

CR (Class Representative)

- Everything a Student can do
- Can submit tickets on behalf of the whole class
- Class and section fields auto-fill from their profile
- Can attach multiple student names to a ticket

Staff

- Can add internal notes, change ticket status, reply in thread, and escalate to Dept Admin if needed
- Can add internal notes, change ticket status, reply in thread
- Gets notifications for assigned tickets and replies

Dept Admin (HOD — Head of Department)

- Sees all department-specific tickets for their own department
- Can reassign tickets within the department
- Can view department-level stats and dashboard
- Cannot see other departments' tickets
- Receives escalated tickets when staff misses deadlines
- Can escalate tickets to Campus Admin when the issue is beyond department authority
- Can assign or reassign tickets to any staff member within the department

Academic Section

- Handles tickets routed to Academic (grades, registration, transcripts, exam schedules)
- Same staff capabilities as CIT but restricted to academic categories

Financial Team

- Handles tickets routed to Financial/Fee (payments, scholarships, refunds)
- Same staff capabilities but restricted to financial categories

Campus Admin (System Admin)

- Manages routing rules (add, edit, delete)
- Sets SLA deadlines per ticket type

- Manages users and roles
- Configures notification email templates
- Sees all tickets across all departments
- Handles tickets escalated from Dept Admins — reviews, escalates further, or resolves
- Exports data, views all reports

2.4 Tech Stack

Layer	Technology
Frontend	React (single-page app, embedded in existing portal)
Backend	Django REST Framework (Python)
Database	PostgreSQL
Authentication	Existing campus login system (OAuth or LDAP-based)
Email	Campus email server (SMTP)
Deployment	Docker containers, CI/CD pipeline
Version Control	Git (feature branches → CIT team review → merge)

2.5 Design and Implementation Constraints

1. Module, not standalone — All URLs, navigation, and styling must fit within CIT's existing system.
2. No duplicate authentication — Must reuse the existing campus login.
3. Branch review workflow — We work on feature branches. Every merge requires CIT team review.
4. No hardcoded secrets — Passwords, API keys, and SMTP credentials must come from environment variables.
5. Mobile-friendly — The UI must work on phone browsers.
6. Attachment size limit — File uploads capped at 10 MB per attachment.

2.6 Assumptions and Dependencies

- CIT's existing system exposes a REST API for user authentication and profile data.
- The campus email server is operational and we have SMTP credentials.

- The PostgreSQL database is provided and managed by CIT (we get a dedicated schema).
- Students already have accounts in the campus system; we do not need a registration flow.
- The existing system's UI has a consistent CSS framework (Bootstrap).
- CIT will provide a staging environment for testing before production deployment.
- Routing rules (R01–R08 from backlog) are the initial configuration.

3. System Features and Requirements

3.1. Ticket Creation and Management

Students and CRs can submit support tickets through a categorized form. Each ticket is automatically assigned a unique ID, categorized, and routed to the appropriate department and staff member. Users can view a list of their tickets and open individual conversation threads to see the full message history. After a ticket is closed, students and CRs can reopen it within 30 days if the issue persists.

Priority: High

Backlog Stories: TKT-001, TKT-002, TKT-003, TKT-004, TKT-006

Sprint: 1, 2

Ticket Lifecycle Flowchart — CIT Support System

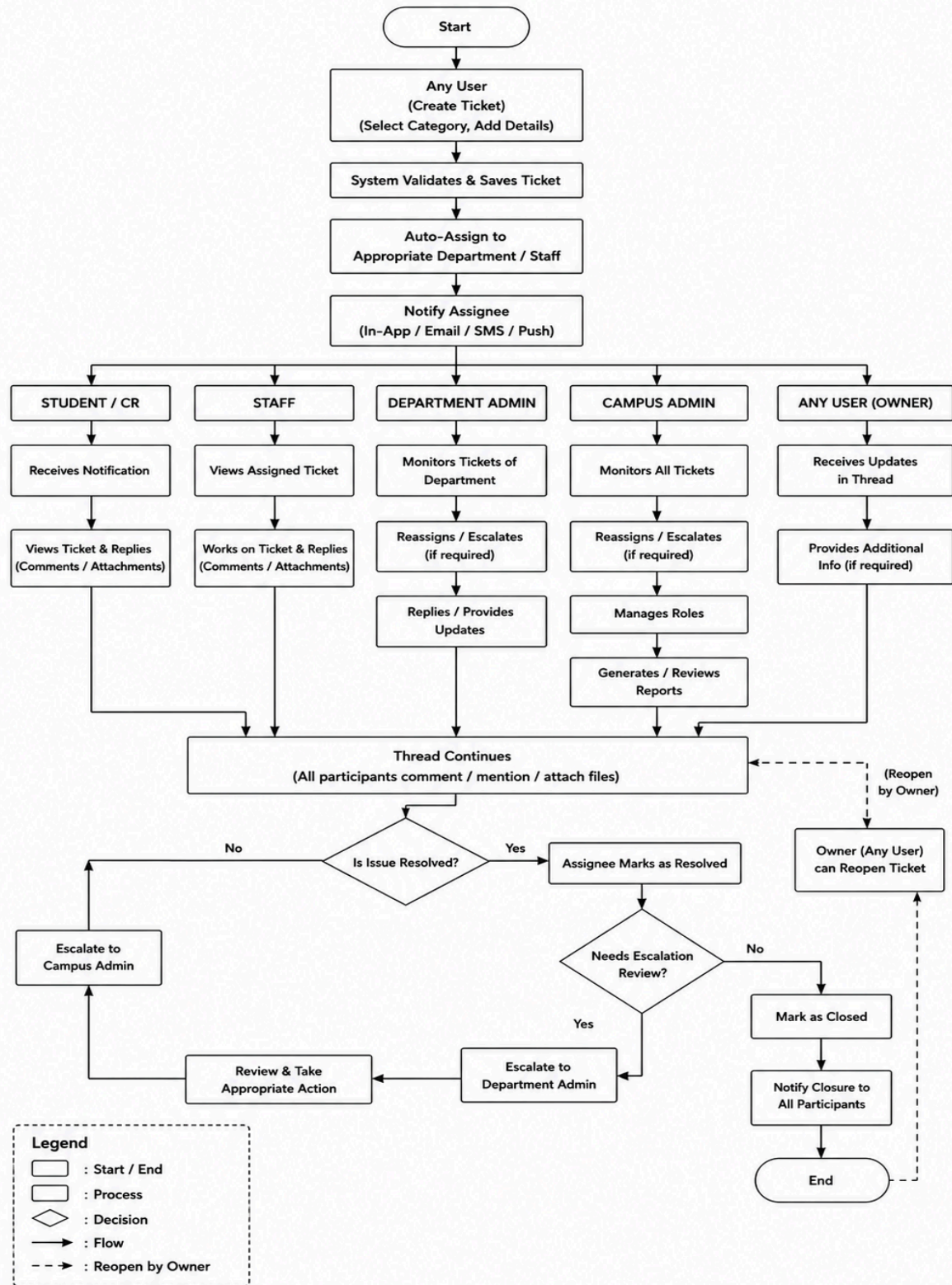


Figure 3: Ticket Lifecycle Flowchart

Stimulus / Response Sequences

Stimulus: Student/CR submits a ticket via the categorized form

Response: System generates a unique Ticket ID, stores the ticket with status 'Open', and notifies the assigned department

Stimulus: Student attaches files up to 10MB

Response: System validates file type and size, uploads file, and links it to the ticket

Stimulus: Student views their submitted tickets

Response: System displays a list of the student's own tickets with status, priority, and last update time

Stimulus: Student replies to their ticket

Response: System appends the reply to the ticket thread and notifies assigned staff

Stimulus: Student reopens a closed ticket (within 30 days)

Response: System changes ticket status from 'Closed' to 'Reopened' and reassigns to the original handler

Stimulus: CR submits a class-level ticket

Response: System auto-fills class/section from CR profile, flags ticket as 'Class-Level', and routes to appropriate department

Actor	Action	System Response
Anyone	Fills ticket form and submits	Creates ticket with ID TKT-XXXXX, shows confirmation, routes to correct team
Student	Opens ticket list	Shows paginated list with ID, title, status, last updated
Student	Clicks on a ticket	Opens thread view with messages in chronological order
Student	Types a reply and attaches a file	Adds reply to thread, notifies assigned staff
Student	Clicks "Close Ticket"	Changes status to Closed, logs the action
Student	Reopens within 30 days	Changes status back to Open, adds a note

CR	Submits ticket for whole class	Includes class/section info, marks ticket as class-level
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Functional Requirements

ID	Requirement
FT-001	The system shall provide a ticket submission form with fields: title (required), description (required), category (dropdown), department (dropdown for department-specific), priority (defaults to Medium), and file attachment
FT-002	The system shall generate a unique ticket ID in the format TKT-XXXXX
FT-003	The system shall display a confirmation screen with the ticket ID immediately after submission
FT-004	The system shall show a paginated list of tickets (20 per page) with columns: ID, title, status, category, last updated
FT-005	The system shall allow sorting by date, status, and priority in the ticket list
FT-006	The system shall allow filtering by status (Open, In Progress, Resolved, Closed) and category
FT-007	The system shall display the full thread for a ticket with all messages in chronological order
FT-008	The system shall hide internal notes from the ticket creator (student/CR)
FT-009	The system shall allow users to reply to a ticket with text and optional file attachments (max 10 MB)
FT-010	The system shall allow the ticket creator to close the ticket when the issue is resolved
FT-011	The system shall allow the ticket creator to reopen a closed ticket within 30 days of closing
FT-012	The system shall log all status changes (who did what and when) visible in the thread

3.2. Ticket Routing and Assignment

When a ticket is submitted, the system automatically categorizes it based on the selected category and department, then assigns it to the appropriate staff member within that department using predefined routing rules. This auto-assignment ensures tickets reach the right handler without manual triage. Routing rules can be configured by a Campus Admin.

Priority: High

Backlog Stories: TKT-007, TKT-008, TKT-009, TKT-010, TKT-011, TKT-012

Sprint: 2, 3, 4

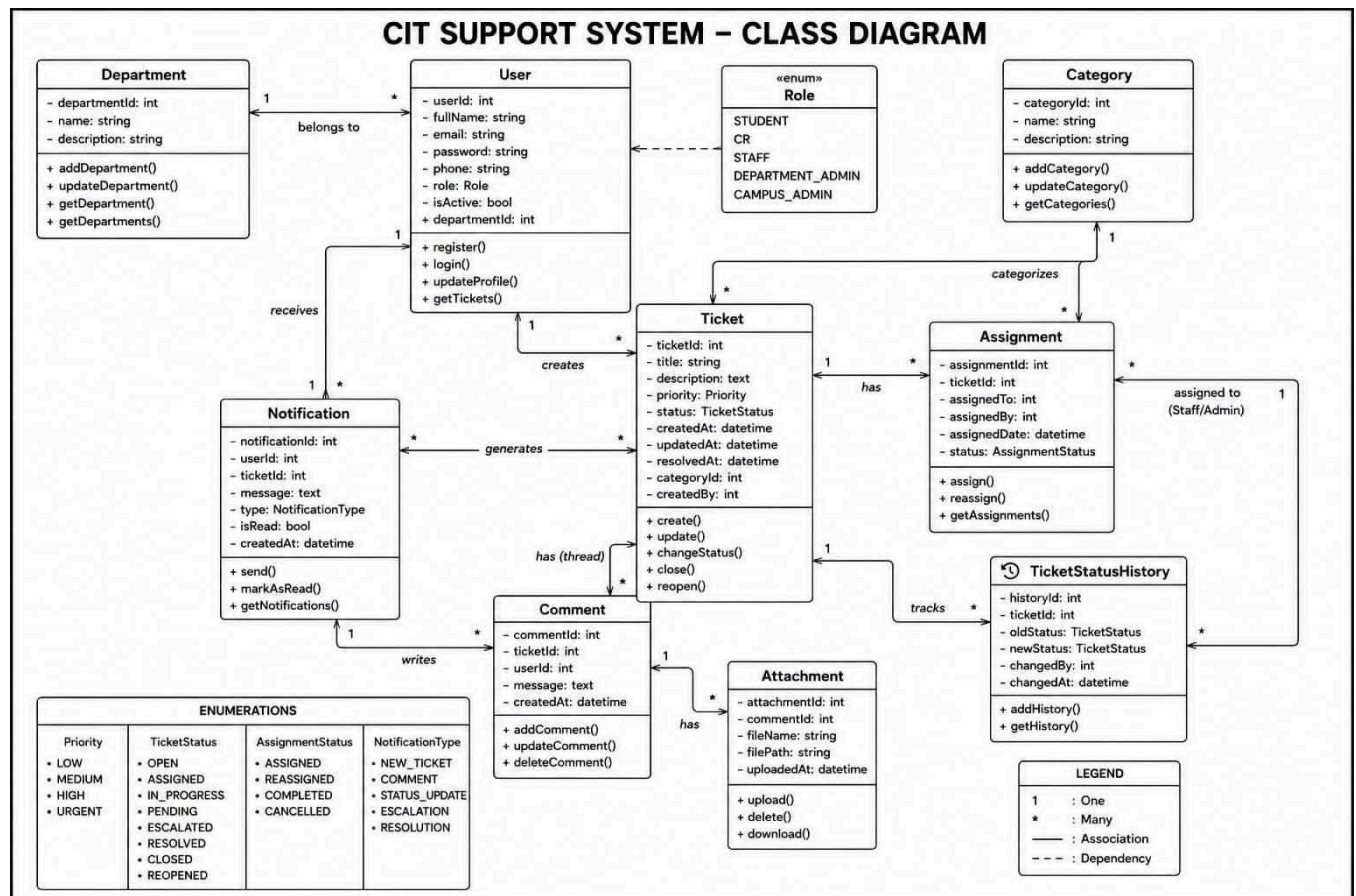


Figure 4: Class Diagram — Ticket Routing and Assignment

Stimulus / Response Sequences

Stimulus: Ticket is submitted with category 'CIT Issue'

Response: System applies routing rule R01: assigns ticket to CIT department staff with round-robin distribution

Stimulus: Ticket is submitted with category 'Academic' (grades, registration, exam)

Response: System applies routing rule R02: assigns ticket to Academic section staff

Stimulus: Ticket is submitted with category 'Financial/Fee'

Response: System applies routing rule R03: assigns ticket to Financial Team staff

Stimulus: Ticket is submitted with category 'HOD/Department'

Response: System applies routing rule R04: routes directly to the HOD of the student's department

Stimulus: Ticket is submitted with category 'Library'

Response: System applies routing rule R05: assigns ticket to Library staff

Stimulus: Ticket is submitted with urgency 'High' or 'Critical'

Response: System applies routing rule R06: assigns with priority flag, shortens SLA deadline, and notifies Dept Admin

Stimulus: Ticket is submitted for a class-level issue by a CR

Response: System applies routing rule R07: routes to the HOD of the relevant department with class-level flag

Stimulus: Ticket category does not match any predefined routing rule

Response: System applies routing rule R08: routes ticket to Campus Admin for manual categorization and assignment

Stimulus: Staff member is unavailable or at max capacity

Response: System applies fallback assignment: routes to the next available staff in the department pool

Stimulus: Staff reassigns a ticket to a different staff member

Response: System logs the reassignment, updates the assigned_to field, and notifies the new assignee

Stimulus: Dept Admin (HOD) manually reassigns a ticket within the department

Response: System updates assignment, adds internal note about reassignment, and notifies both old and new assignees

Actor	Action	System Response
System	User submits "Internet" category ticket	Routes to CIT team, sends notification

System	User submits "Academic" category ticket	Routes to Academic Section, CCs concerned HOD
System	User submits "Financial" category ticket	Routes to Financial Team, marks scholarship queries as priority
System	User submits "Department-specific" ticket	Routes to respective HOD based on department field
Admin	Adds a new routing rule	Rule goes live immediately, history logged
System	Assigned person is on leave	Falls back to team lead, adds note in thread

Functional Requirements

ID	Requirement
FT-013	The system shall route tickets based on category and department mapping: the selected category determines the ticket type, and the department field determines which department handles it. Dept Admin assigns to appropriate staff.
FT-014	The system shall apply routing rules R01–R08 from the product backlog as the default configuration
FT-015	The system shall provide an admin UI to add, edit, delete, and reorder routing rules without requiring code changes
FT-016	The system shall log all routing rule changes with timestamp and admin name
FT-017	The system shall support keyword-based matching within ticket titles/descriptions as an additional routing condition
FT-018	The system shall check staff leave status before assigning and fall back to the team lead or department head if the primary assignee is unavailable
FT-019	The system shall add a note to the ticket thread whenever re-assignment happens due to leave

3.3. Notifications

Users get notified via email and in-app notifications when tickets are assigned, updated, or nearing deadlines. Admins can customize notification templates.

Priority: High

Backlog Stories: TKT-013, TKT-014, TKT-015, TKT-016, TKT-017

Sprint: 2, 3, 4

Stimulus / Response Sequences

Stimulus: New ticket is assigned to a staff member

Response: System sends email and in-app notification to the assigned staff with ticket details and priority

Stimulus: Staff adds a reply to the ticket thread

Response: System notifies the ticket submitter (student/CR) via email and in-app notification

Stimulus: Staff adds an internal note

Response: System does NOT notify the student; only department staff and admins can see the note

Stimulus: Ticket deadline is approaching (80% of SLA elapsed)

Response: System sends a warning notification to assigned staff and CC to Dept Admin

Stimulus: Deadline is missed

Response: System escalates the ticket to Dept Admin and sends escalation alert

Stimulus: User reopens a ticket

Response: System notifies the original assigned staff and the Dept Admin

Stimulus: Admin configures notification templates

Response: System saves the template and applies it to future notifications

Actor	Action	System Response
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Staff	Ticket assigned to them	Email sent within 1 minute, in-app badge increments
Student	Their ticket status changes	Email sent (Assigned, Resolved, Closed), in-app notification
Staff	Someone replies to their ticket	Notification within 30 seconds, badge count updates
Admin	Edits notification template	Preview shown before save, changes apply to new emails only
Staff	Ticket nearing deadline	Warning alert 1 hour before deadline (email + in-app)

Functional Requirements

ID	Requirement
FT-020	The system shall send an email notification to staff within 1 minute when a ticket is assigned to them
FT-021	The system shall send an email notification to the ticket creator when their ticket status changes to Assigned, Resolved, or Closed
FT-022	The system shall display an in-app notification bell icon with an unread count for all users
FT-023	The system shall send an in-app notification when someone replies to a ticket the user is involved in
FT-024	The system shall allow users to click an in-app notification to navigate directly to the relevant ticket
FT-025	The system shall allow admins to edit email notification subject lines and body templates using placeholders like {ticket_id}, {title}, {status}
FT-026	The system shall show a preview of the email before saving template changes
FT-027	The system shall send a warning notification 1 hour before a ticket's response deadline (if configured)
FT-028	The system shall provide an opt-out toggle for user who do not want email notifications

3.4. Escalation and Deadlines

The system implements a multi-level escalation chain to ensure no ticket goes unattended. When a staff member misses a response deadline, the ticket is automatically escalated to the Dept Admin (HOD). The Dept Admin can then assign it to another staff member within the department, resolve it directly, or escalate it further to the Campus Admin if the issue is beyond departmental authority. The Campus Admin reviews escalated tickets and takes appropriate action. Campus Admins can also set response-time and resolution-time deadlines for each ticket category.

Priority: High

Backlog Stories: TKT-018, TKT-019, TKT-020

Sprint: 3, 4, 5

Stimulus / Response Sequences

Stimulus: Staff fails to respond within SLA deadline

Response: System escalates ticket to Dept Admin (HOD) with status changed to 'Escalated-Level1'

Stimulus: Dept Admin fails to resolve escalated ticket within SLA

Response: System escalates ticket to Campus Admin with status changed to 'Escalated-Level2'

Stimulus: Ticket priority is 'Critical' and remains unassigned for >30 minutes

Response: System auto-assigns to Dept Admin directly and sends escalation alert

Stimulus: Campus Admin receives escalated ticket

Response: System displays escalation history, sets status to 'Admin-Review', and allows reassignment to any department

Stimulus: Dept Admin resolves an escalated ticket

Response: System changes status to 'Resolved', logs resolution time, and notifies the submitter

Stimulus: Ticket is escalated beyond department authority

Response: System flags ticket as 'Cross-Department' and Campus Admin reassigns to the appropriate department

Actor	Action	System Response
Admin	Sets deadline: CIT tickets → 4h response	SLA timer starts on ticket creation

System	Staff deadline passed with no response	Auto-escalates ticket to Dept Admin (HOD), flags as overdue
Admin	Views dashboard	Sees missed deadline %, filterable by team
HOD	Downloads report	Gets CSV with deadline compliance data
Dept Admin	Receives escalated ticket from staff	Reviews ticket, assigns to another staff or escalates to Campus Admin
Dept Admin	Escalates to Campus Admin	Ticket moved to Campus Admin queue, notification sent, escalation logged
Campus Admin	Receives escalated ticket from Dept Admin	Reviews ticket, takes action, or resolves directly
All User	Issue not resolved after ticket is closed	Reopens ticket within 30 days, status reverts to Open, notification sent

Functional Requirements

ID	Requirement
FT-036	The system shall allow admins to configure response-time and resolution-time deadlines per ticket category and priority level
FT-037	The system shall track SLA time only during working hours (configurable by admin)
FT-038	The system shall automatically escalate a ticket to the HOD or CIT head if the response deadline is exceeded and no action has been taken
FT-039	The system shall display a visible "Overdue" flag on escalated tickets
FT-040	The system shall notify the original assignee when a ticket is escalated from them

FT-041	The system shall provide a dashboard view showing deadline compliance metrics (missed % per team, per category)
FT-042	The system shall allow downloading the deadline compliance report as CSV
FT-029	The system shall implement a multi-level escalation chain: Staff → Dept Admin (HOD) → Campus Admin
FT-030	The system shall allow Dept Admins to reassign escalated tickets to any staff member within their department
FT-031	The system shall allow Dept Admins to escalate tickets to Campus Admin when the issue is beyond departmental authority
FT-032	The system shall allow Campus Admins to review, resolve, or further escalate tickets received from Dept Admins
FT-033	The system shall allow ticket owner to reopen a closed ticket within 30 days if the issue is not resolved
FT-034	The system shall send notifications at each escalation level to all involved parties
FT-035	The system shall maintain a complete escalation history log visible in the ticket thread

3.5. User and Role Management

Admins can add, edit, delete users and assign roles. CRs have special privileges for class-level tickets. HODs see only their department's tickets.

Priority: High

Backlog Stories: TKT-021, TKT-022, TKT-023

Sprint: 1, 3

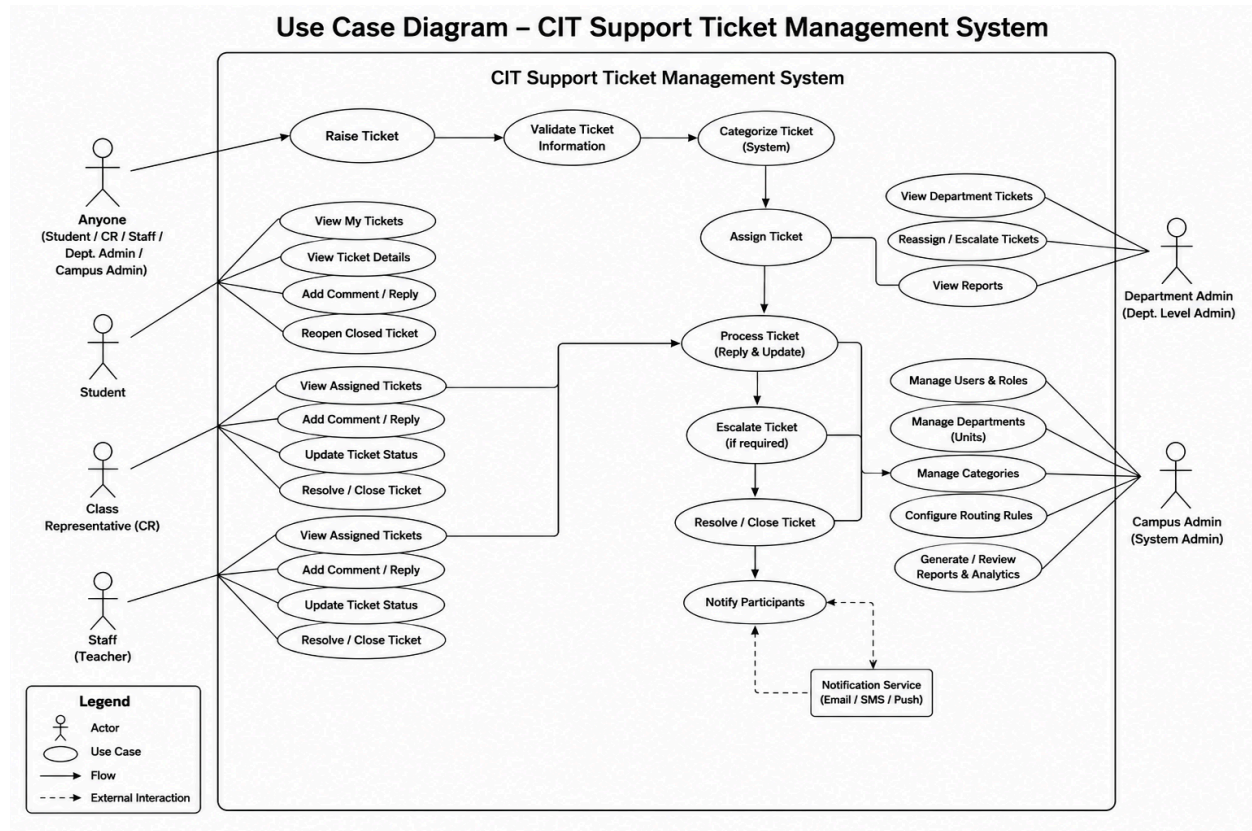


Figure 5: Use Case Diagram — User Roles and Permissions

Gantt Chart : CIT Support Ticket Management Module

GANTT CHART - CIT SUPPORT TICKET MANAGEMENT MODULE

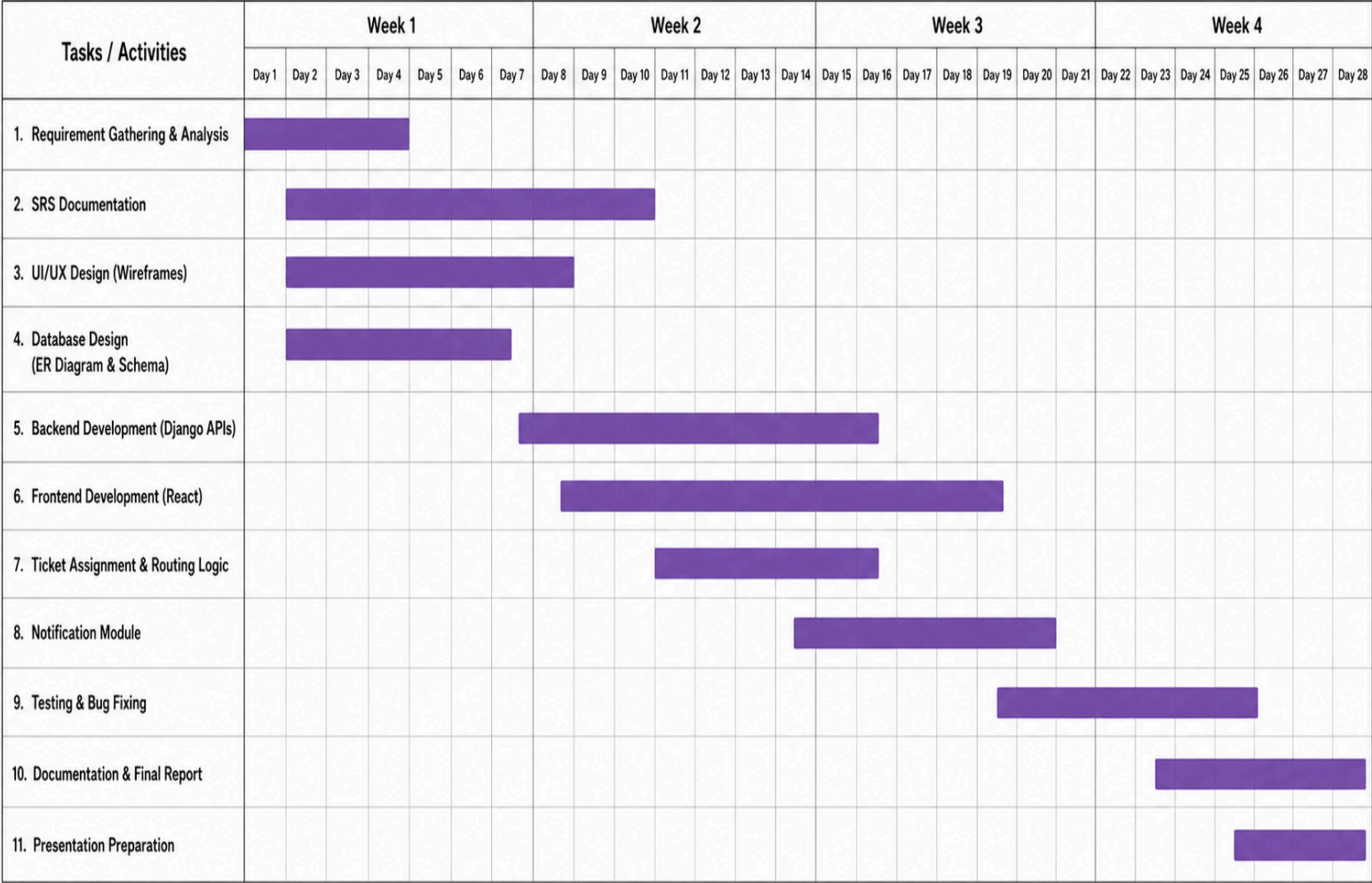


Figure 6: Gantt Chart

Stimulus / Response Sequences

Actor	Action	System Response
Admin	Adds new user with role "CIT Staff"	User appears in CIT team, can be assigned tickets

Admin	Bulk imports students from Excel	All users created with "Student" role
CR	Submits a class ticket	Class/section auto-filled, ticket marked "For whole class"
HOD	Opens ticket list	Sees only tickets from their department
HOD	Reassigns a ticket	Ticket assigned to another staff in same department

Note : some roles may be added with requirements.

Functional Requirements

ID	Requirement
FT-036	The system shall provide an admin interface to add, edit, and deactivate users with roles: Student, CR, Staff, Dept Admin, Campus Admin
FT-037	The system shall support bulk user import from an Excel file (columns: name, email, department, section, role)
FT-038	The system shall allow CRs to submit tickets marked as "For whole class" with auto-filled class and section from the CR's profile
FT-039	The system shall allow HODs to view and filter tickets belonging only to their own department
FT-040	The system shall allow HODs to reassign tickets to any staff member within their department
FT-041	The system shall enforce role-based access: students see only their own tickets, HODs see department tickets, admins see everything

3.6. Search

Users can search through past tickets by keyword, ticket ID, or category. Results are sorted by relevance.

Priority: Medium

Backlog Stories: TKT-024

Sprint: 5

Stimulus / Response Sequences

Actor	Action	System Response
Any user	Types keyword in search bar	Returns matching tickets sorted by relevance
Any user	Filters by date range	Results narrowed to date range
Any user	Filters by status	Only tickets matching status shown

Functional Requirements

ID	Requirement
FT-042	The system shall provide a full-text search across ticket ID, title, description, and message content
FT-043	The system shall display search results sorted by relevance (best match first)
FT-044	The system shall allow filtering search results by date range, status, and category

3.7. Reports and Dashboard

HODs and admins get visual dashboards showing ticket statistics, staff performance, and exportable data.

Priority: Medium (High for dashboard)

Backlog Stories: TKT-025, TKT-026, TKT-027

Sprint: 5, 6

Stimulus / Response Sequences

Actor	Action	System Response
HOD	Opens dashboard	Shows pie chart (open vs closed), bar chart (by category), avg resolution time
Admin	Clicks "Export to Excel"	Prompts for date range and columns, downloads file
Admin	Opens staff metrics	Shows tickets handled per person, avg response time, weekly trends

Functional Requirements

ID	Requirement
FT-045	The system shall display a dashboard with: pie chart of open vs closed tickets, bar chart of tickets by category, and average resolution time
FT-046	The system shall allow users to export ticket data to Excel with configurable date range and column selection
FT-047	The system shall show per-staff metrics: tickets handled, average response time, and weekly trend for admins and HODs

4. External Interface Requirements

4.1 User Interfaces

- The system is a React single-page application embedded within CIT's existing campus portal
- The UI must match the existing portal's look and feel (CSS variables, font, color scheme provided by CIT)
- Mobile-responsive: all pages must work on screens as small as 360px wide
- The ticket thread view should resemble a messaging app — messages in bubbles, left-aligned for students, right-aligned for staff
- Internal notes display with a yellow background and "Internal Note" label
- The ticket list shows a clean table layout with sortable columns on desktop and a card layout on mobile
- Forms use clear labels, validation errors appear inline below fields
- Loading states (skeleton screens or spinners) for all API calls
- Error messages are plain language, not technical stack traces

4.2 Hardware Interfaces

Standard web browsers (Chrome, Firefox, Safari, Edge — latest 2 major versions). No specialized hardware required. File uploads handled via standard HTML file input.

4.3 Software Interfaces

System	Interface	Direction	Data
CIT Auth System	REST API	Read	User identity, roles, department
CIT Student Database	REST API (or DB view)	Read	Student profiles, section, batch
Campus Email Server	SMTP	Write	Sends notification emails
CIT Portal (parent app)	JavaScript embed	Both	Navigation events, auth token sharing
PostgreSQL	Database connection	Both	All ticket data

4.4 Communication Interfaces

All frontend-to-backend communication is via REST API over HTTPS. API base URL: /api/tickets/ (namespaced under existing portal's API). Authentication token passed via HTTP header. Request and response bodies in JSON format. File uploads use multipart/form-data.

API responses follow a consistent format:

```
{  
  "success": true,  
  "data": { ... },  
  "error": null  
}
```

Paginated responses include count, next, previous, and results fields (standard Django REST pagination). WebSocket connections for real-time in-app notifications (optional; fallback to polling if WS not feasible).

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-001	The system shall load the ticket list page in under 2 seconds for a user with 100+ tickets
NFR-002	The system shall send email notifications within 1 minute of the triggering event
NFR-003	The system shall support at least 50 concurrent users without degradation
NFR-004	The ticket submission form shall submit and display confirmation within 3 seconds
NFR-005	Search queries shall return results within 3 seconds for the full ticket database

5.2 Security

ID	Requirement
NFR-006	The system shall enforce role-based access control (RBAC) — users can only see and act on tickets they are authorized for
NFR-007	No passwords, API keys, or SMTP credentials shall be hardcoded in the codebase
NFR-008	API endpoints shall validate authentication on every request. Unauthenticated requests return 401
NFR-009	File uploads shall be scanned for malware (CIT to provide antivirus integration if available)
NFR-010	Internal notes shall never be returned by any API endpoint that a student/CR can access
NFR-011	All API traffic shall be over HTTPS only

5.3 Reliability

ID	Requirement
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NFR-012	The system shall target 99% uptime during campus working hours (8 AM – 6 PM, Sun–Thu)
NFR-013	In case of database failure, the system shall show a friendly error page rather than a stack trace
NFR-014	Ticket submissions shall be atomic — if the submission fails partway, no partial data is saved
NFR-015	The system shall handle email server failures gracefully (log the error, do not crash, retry on next trigger)

5.4 Usability

ID	Requirement
NFR-016	The ticket list and thread view shall work on mobile browsers without horizontal scrolling
NFR-017	The ticket thread view shall follow a chat-like layout (messages in bubbles, newest at bottom)
NFR-018	Forms shall show inline validation errors (not popup alerts)
NFR-019	The system shall provide a search bar accessible from all pages
NFR-020	The system shall use consistent terminology across all screens

Appendix A: Sample Rules, Roles and Glossary

Term	Definition
CR	Class Representative — a student who represents their batch/section
CIT	Computer and Information Technology — the campus IT department
HOD	Head of Department — oversees an academic department
SLA	Service Level Agreement — target response/resolution times
Ticket	A support request record with ID, thread, status, and assignment
Internal Note	A staff-only message not visible to the ticket creator
Routing Rule	A configuration that maps ticket categories to assignee teams

Appendix B: Sample Departments and Roles

Department	Code	HOD Role
Civil Engineering	CIV	hod.civil@pulchowk.edu
Electrical Engineering	ELE	hod.electrical@pulchowk.edu
Computer Engineering	COM	hod.computer@pulchowk.edu
Mechanical Engineering	MEC	hod.mechanical@pulchowk.edu
Architecture	ARC	hod.architecture@pulchowk.edu
Applied Sciences	APP	hod.appsci@pulchowk.edu

Appendix C: Sample Default Routing Rules

Category	Assign To	Response Time	Escalate To
Internet / Network / Technical	CIT Team	4 hours	CIT Head
Hardware / Lab Equipment	CIT (Hardware)	8 hours	CIT Head
Academic	Academic Section	24 hours	Concerned HOD
Financial / Fees	Financial Team	24 hours	Finance Head
Department-specific	Respective HOD	24 hours	Campus Chief
Hostel / Facilities	Facilities Team	24 hours	Admin Head
Library	Library Staff	24 hours	Librarian
General / Other	CIT Triage	48 hours	CIT Head